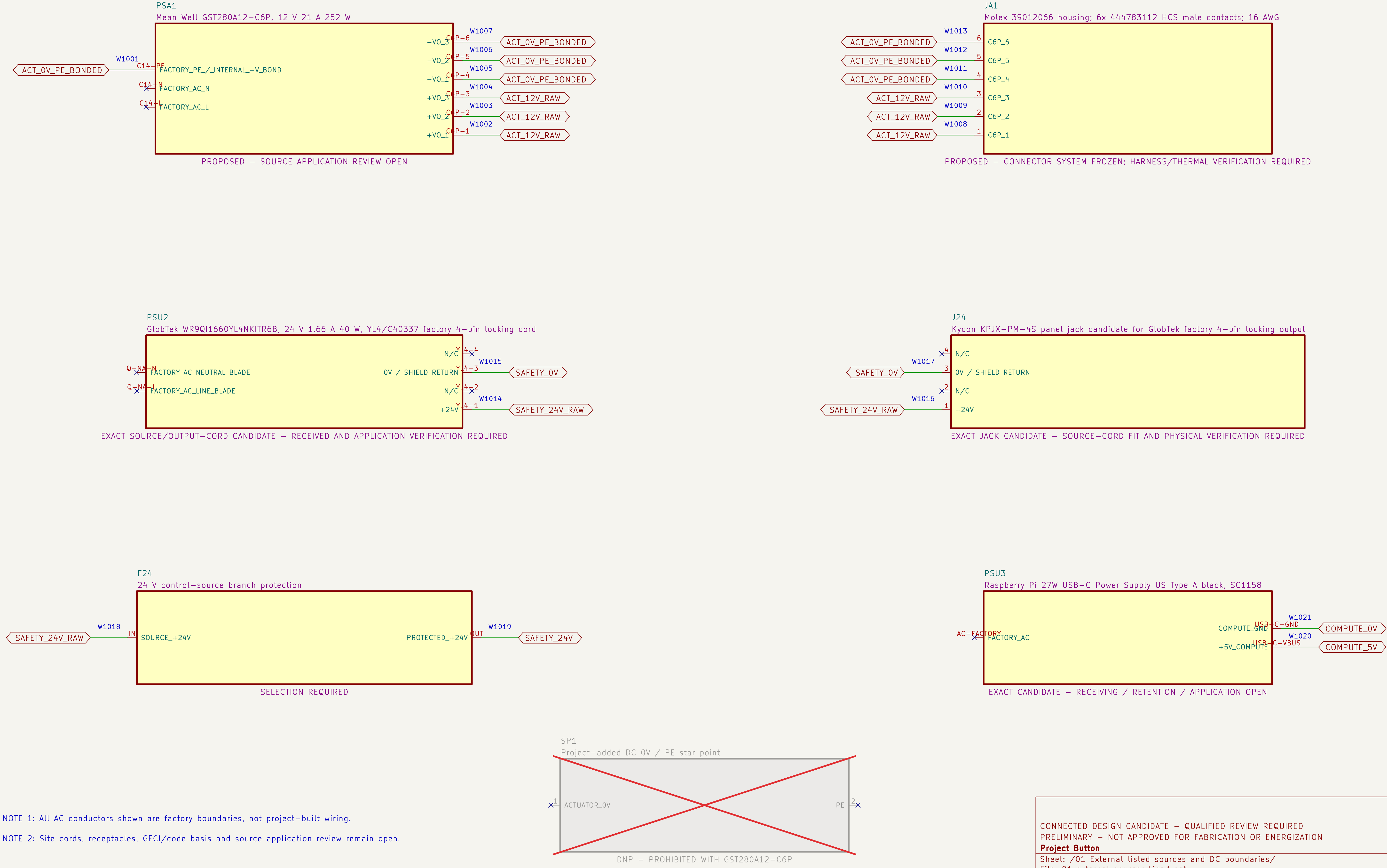
PROJECT BUTTON HR–V0 ELECTRICAL V3 CONNECTED CANDIDATE

Direct dual–channel E–stop inputs, watchdog–gated SR1 supply, separate RESET/ARM, redundant actuator interruption.

01 External listed sources and DC boundaries

Factory−sealed adapters eliminate project−built mains wiring.



NOTE 1: All AC conductors shown are factory boundaries, not project−built wiring.

NOTE 2: Site cords, receptacles, GFCI/code basis and source application review remain open.

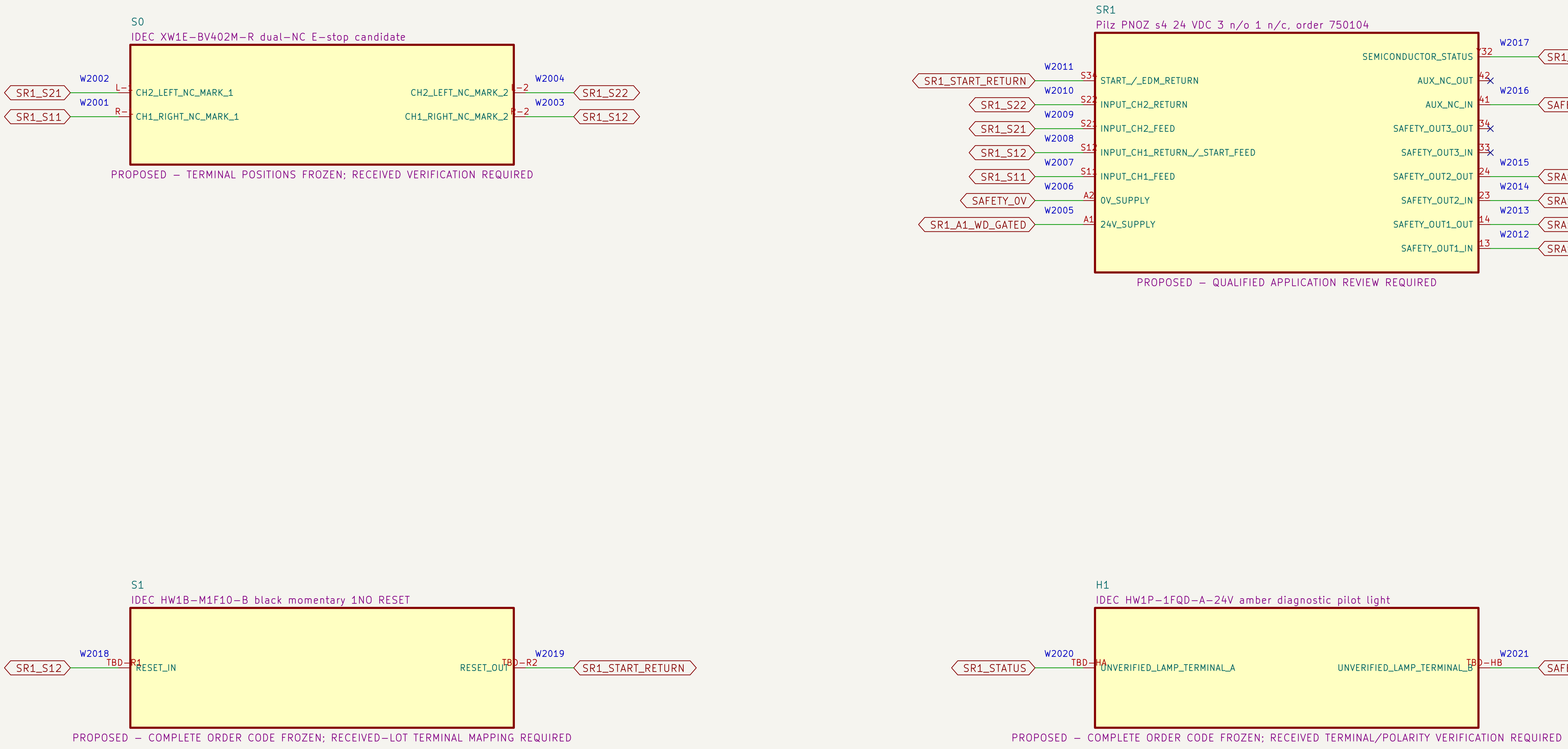
CONNECTED DESIGN CANDIDATE — QUALIFIED REVIEW REQUIRED
PRELIMINARY — NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /01 External listed sources and DC boundaries/
File: 01_external_sources.kicad_sch

Title: PB HR−V0 V3−P1.14 — 01

Size: A3	Date: 2026−08−08	**Rev: V3−P1.14**
KiCad E.D.A. 10.0.5		Id: 2/13

02 Dual-channel E-stop and RESET eligibility

Each SR1 input return contains only one S0 NC contact; RESET cannot energize K1/K2.



NOTE 1: S0 directly completes both SR1 input returns; ordinary KWD terminals are absent from both loops.

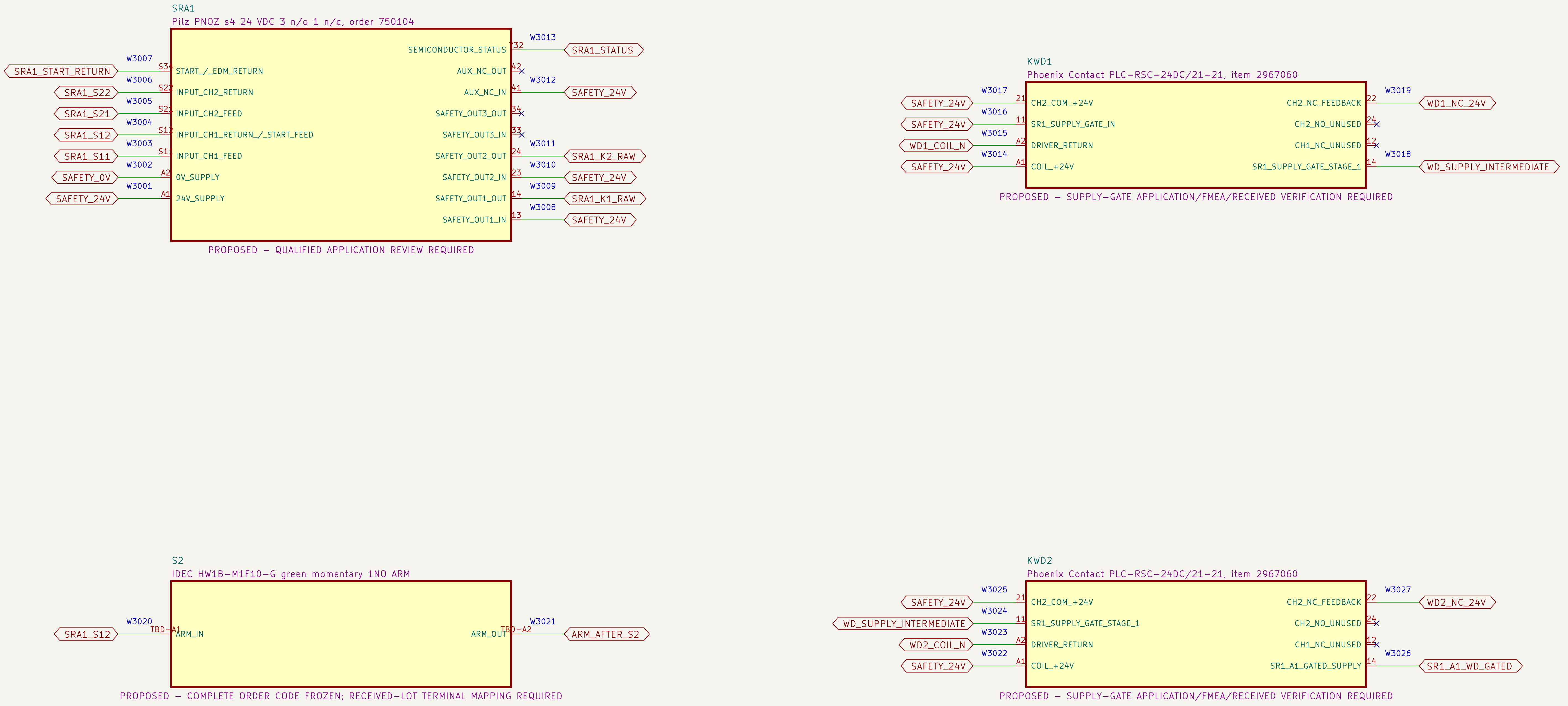
NOTE 2: Heartbeat loss removes SR1 A1 through the separate two-contact supply gate; recovery alone cannot restore the monitored RESET stage.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /02 Dual-channel E-stop and RESET eligibility/
File: 02_estop_eligibility.kicad_sch
Title: PB HR-V0 V3-P1.14 – 02

Size: A3	Date: 2026-08-08	**Rev: V3-P1.14**
KiCad E.D.A. 10.0.5		Id: 3/13

03 Distinct ARM and watchdog-gated SR1 supply

Two ordinary KWD contacts gate SR1 A1; SRA1 still requires SR1 outputs, EDM proof and a new ARM action.



NOTE 1: Required after E-stop/watchdog dropout: cause healthy -> KWD supply restored ->
RESET press/release -> SAFE_READY -> distinct ARM press/release.

NOTE 2: KWD contacts are in series with SR1 A1 only; S0 remains direct in both SR1 input
returns and SRA1 receives both SR1 safety outputs directly.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /03 Distinct ARM and watchdog-gated SR1 supply/
File: 03_arm_watchdog_eligibility.kicad_sch
Title: PB HR-V0 V3-P1.14 – 03
Size: A3 | Date: 2026-08-08 | **Rev: V3-P1.14**
KiCad E.D.A. 10.0.5 | Id: 4/13

PRELIMINARY — NOT APPROVED FOR FABRICATION OR ENERGIZATION

04 Contactor coils, mirror contacts and EDM

K1 and K2 are distinct final elements; their mirror contacts form the monitored restart return.

NOTE 1: SRA1 outputs are separately protected before K1 and K2 coils. FSR1/FSR2 use proposed Phoenix PT 4−HESI item 3211861 holders with D−ST 4 item 3030420 end cover; both fuse links, grouping and coordination remain SELECTION REQUIRED.

NOTE 2: Loaded DC interruption, suppression behavior, mirror−contact use and coordination require qualified review.

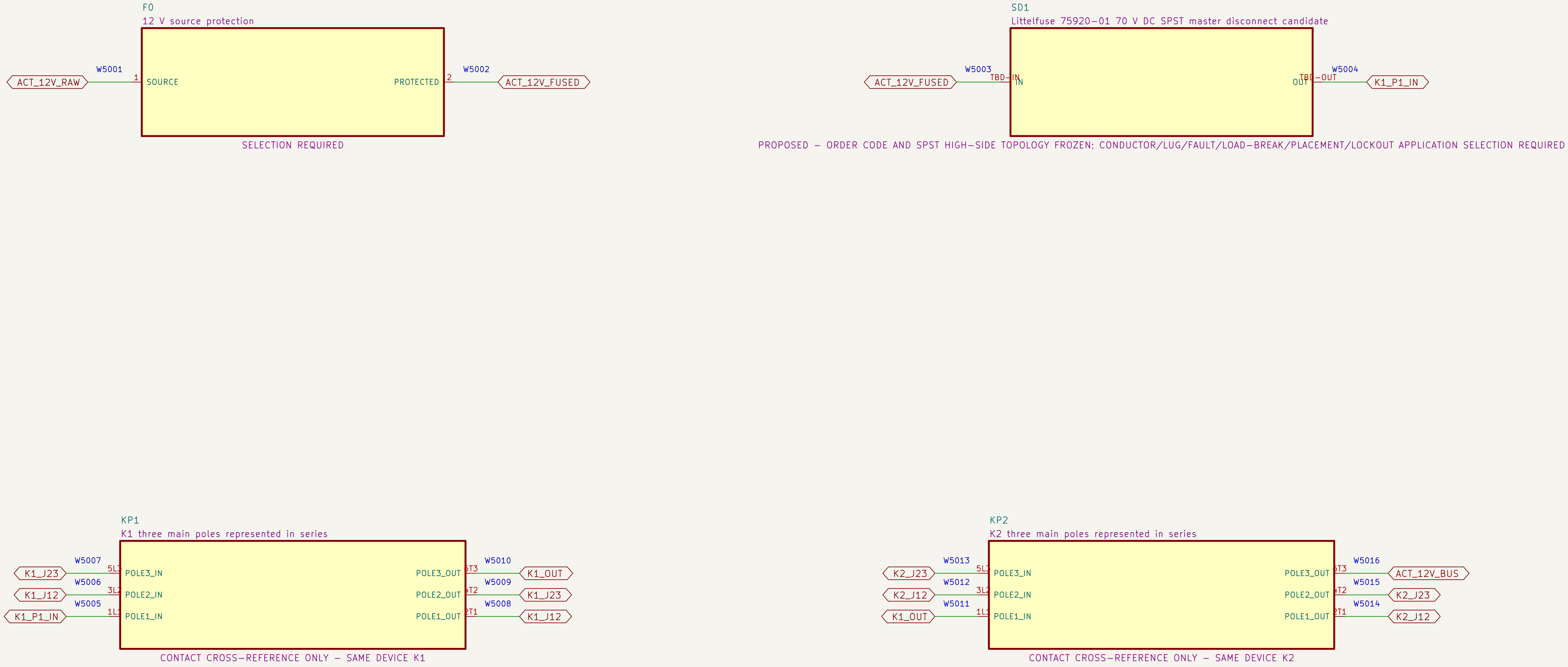
CONNECTED DESIGN CANDIDATE — QUALIFIED REVIEW REQUIRED
PRELIMINARY — NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /04 Contactor coils, mirror contacts and EDM/
File: 04_contactor_edm.kicad_sch

Title: PB HR−V0 V3−P1.14 — 04

Size: A3	Date: 2026−08−08	**Rev: V3−P1.14**
KiCad E.D.A. 10.0.5		Id: 5/13

05 Redundant actuator–power interruption

Source protection, service disconnect and all three poles of K1 then K2 are represented in series.



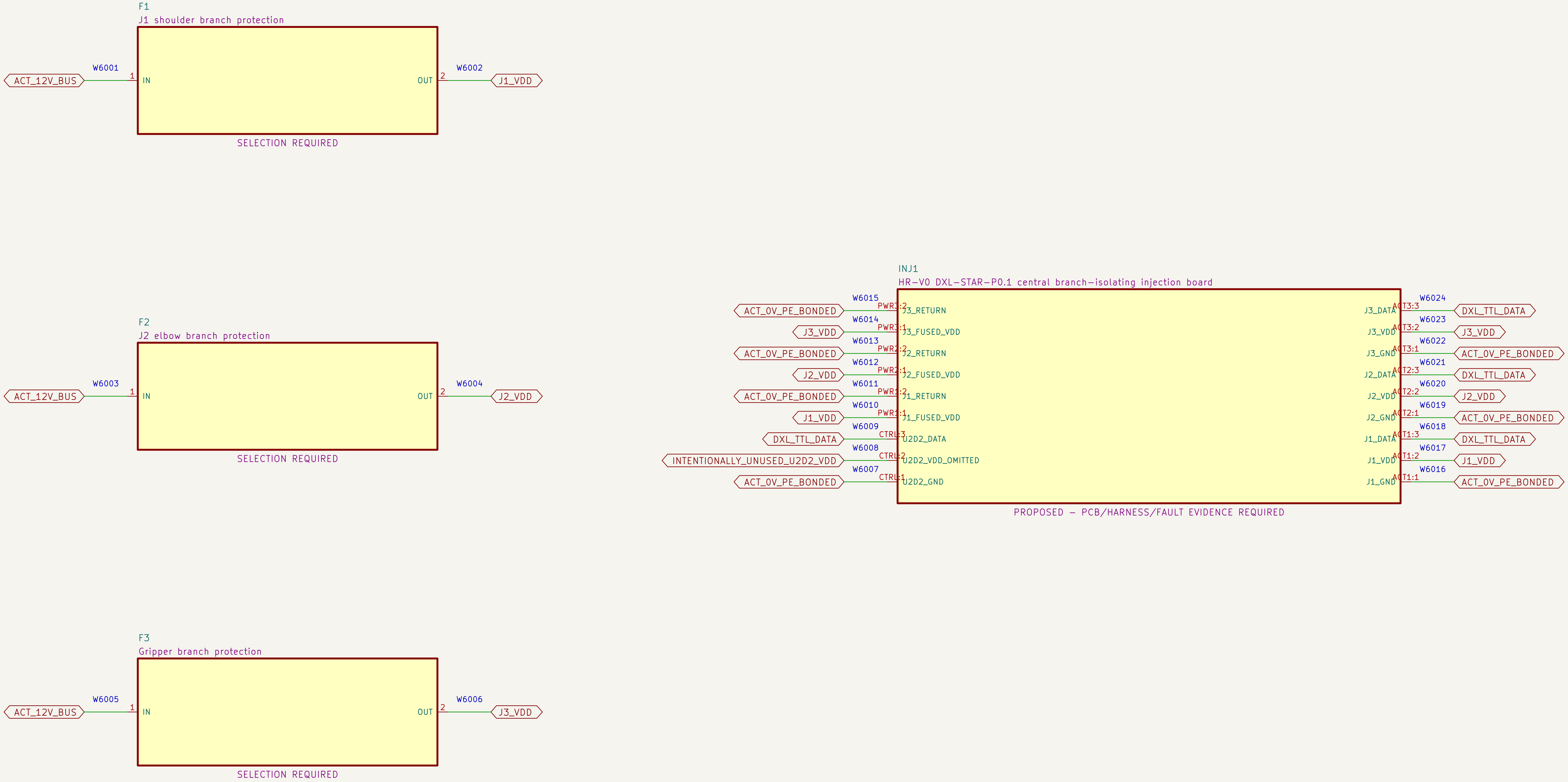
NOTE 1: Series jumpers: K1 2T1->3L2, 4T2->5L3, 6T3->K2 1L1; repeat through K2 to ACT_12V_BUS.

NOTE 2: No fuse, disconnect, conductor, connector or contactor application is released without fault-current evidence.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /05 Redundant actuator-power interruption/
File: 05_actuator_interruption.kicad_sch
Title: PB HR-V0 V3-P1.14 – 05
Size: A3 | Date: 2026-08-08 | **Rev: V3-P1.14**
KiCad E.D.A. 10.0.5 | Id: 6/13

06 Protected actuator branches and central DYNAMIXEL star injection

Each actuator has a separate protected VDD branch; U2D2 pin 2 and inter–actuator VDD paths are omitted.



NOTE 1: INJ1 is a fixed central star: U2D2 pin 2 is unrouted and J1/J2/J3 positive rails never join.

NOTE 2: Exact actuator cables, branch returns, enclosure, continuity, isolation, signal–integrity, pull and no–backfeed tests remain release gates.

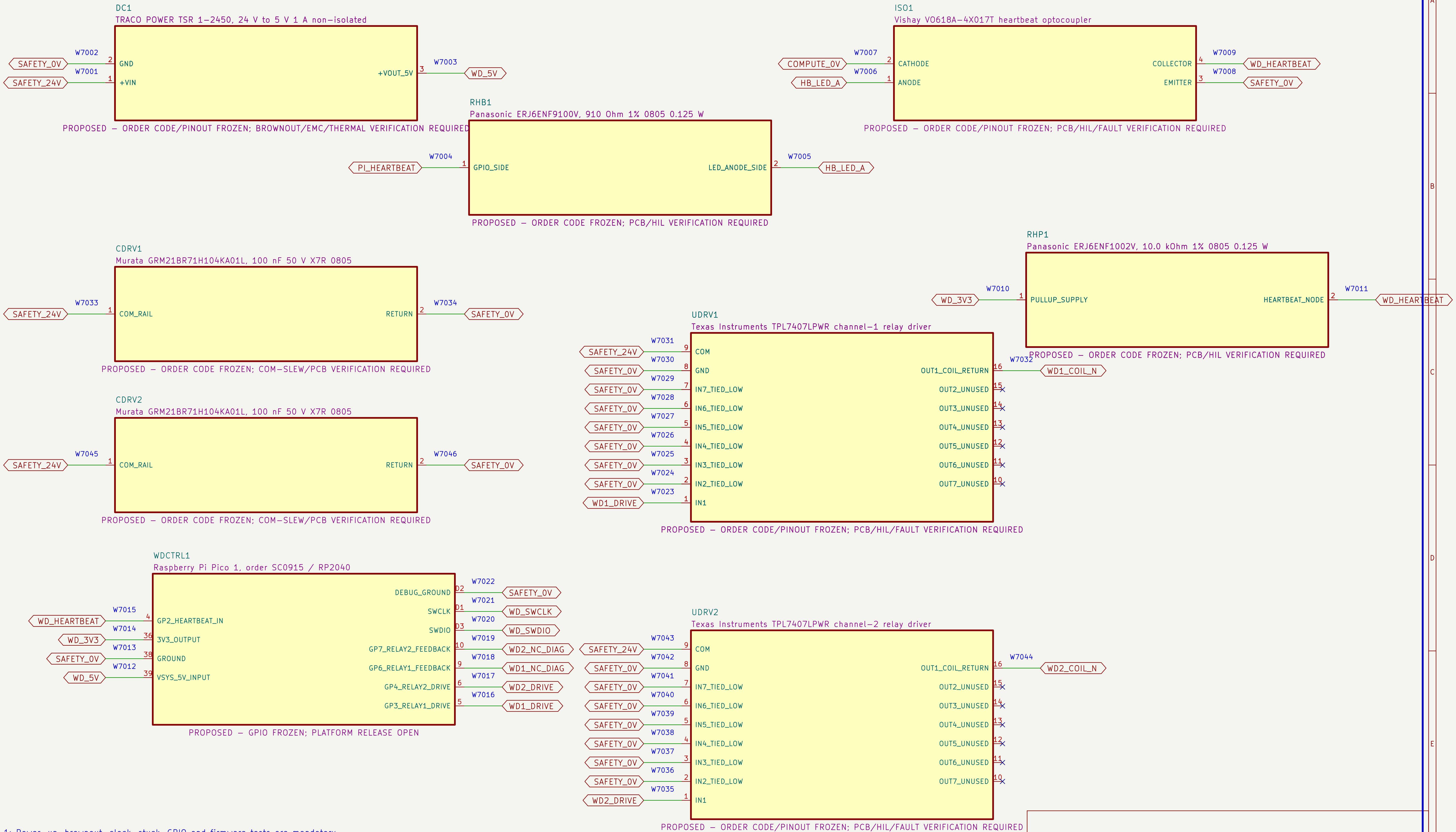
CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /06 Protected actuator branches and central DYNAMIXEL star injection/
File: 06_branches_and_injection.kicad_sch

Title: PB HR–V0 V3–P1.14 – 06

Size: A3	Date: 2026–08–08	**Rev: V3–P1.14**
KiCad E.D.A. 10.0.5		Id: 7/13

07 Independent watchdog power, controller and drivers

The ordinary watchdog controller and drivers receive no safety–integrity credit by assertion.



NOTE 1: Power–up, brownout, clock, stuck–GPIO and firmware tests are mandatory.

NOTE 2: Qualified review decides whether watchdog loss is credited or diagnostic only;

current safety credit is NONE.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION

Project Button

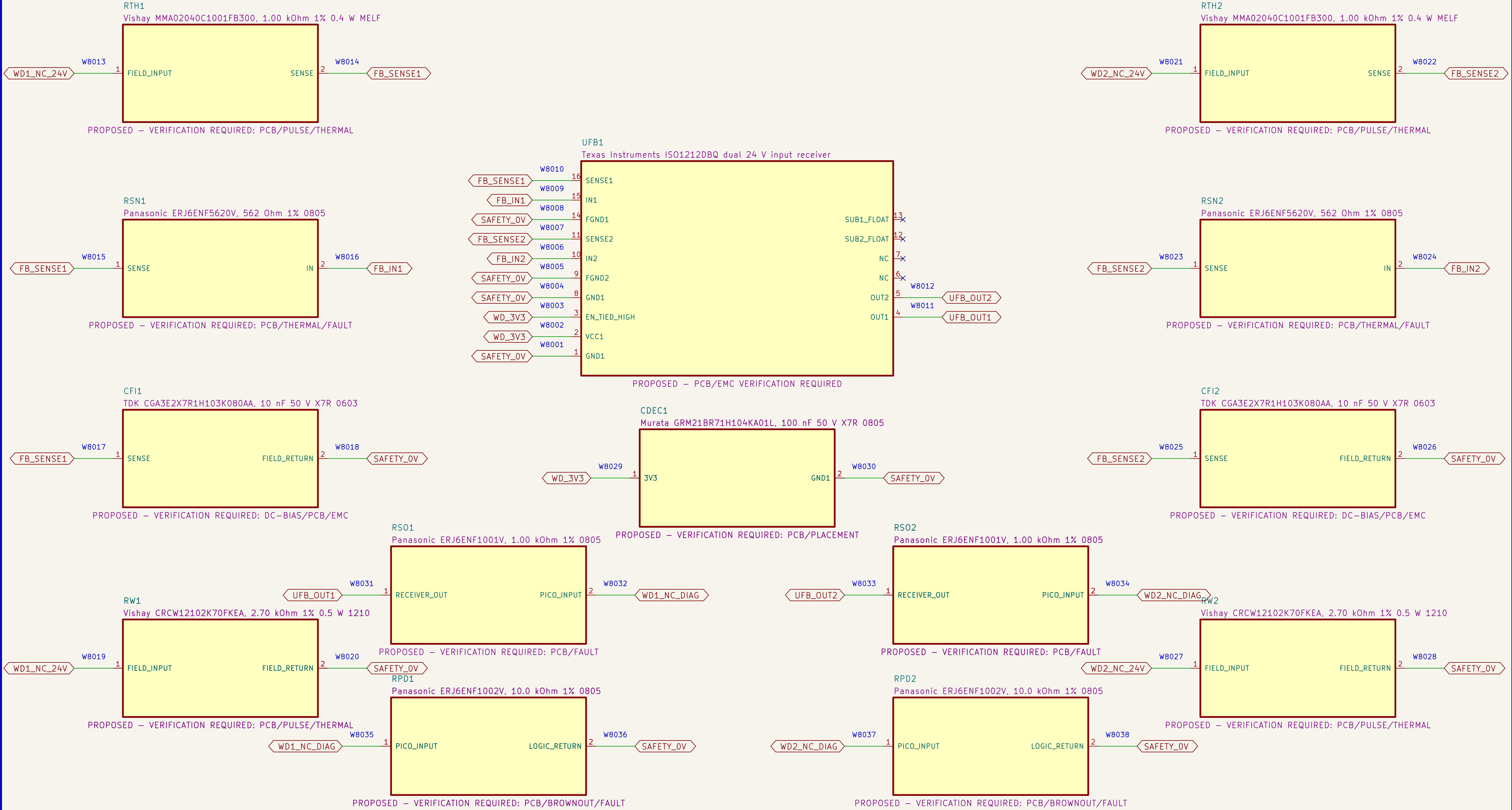
Sheet: /07 Independent watchdog power, controller and drivers/
File: 07_watchdog_control.kicad_sch

Title: PB HR–V0 V3–P1.14 – 07

Size: A3	Date: 2026–08–08	**Rev: V3–P1.14**
KiCad E.D.A. 10.0.5		Id: 8/13

08 Calculated dual–channel 24 V watchdog feedback

ISO1212DBQ converts both relay NC diagnostics to default–low 3.3 V logic; no isolation or safety credit is claimed.



NOTE 1: Type–3 values: RTHR=1.00 kOhm, RSENSE=562 Ohm, CIN=10 nF; calculated wetting load is 2.70 kOhm 1% 0.5 W per channel.

NOTE 2: Both grounds are SAFETY_0V; the ISO1212 barrier is not credited. Passive identities are frozen; PCB, derating, EMC and fault tests remain open.

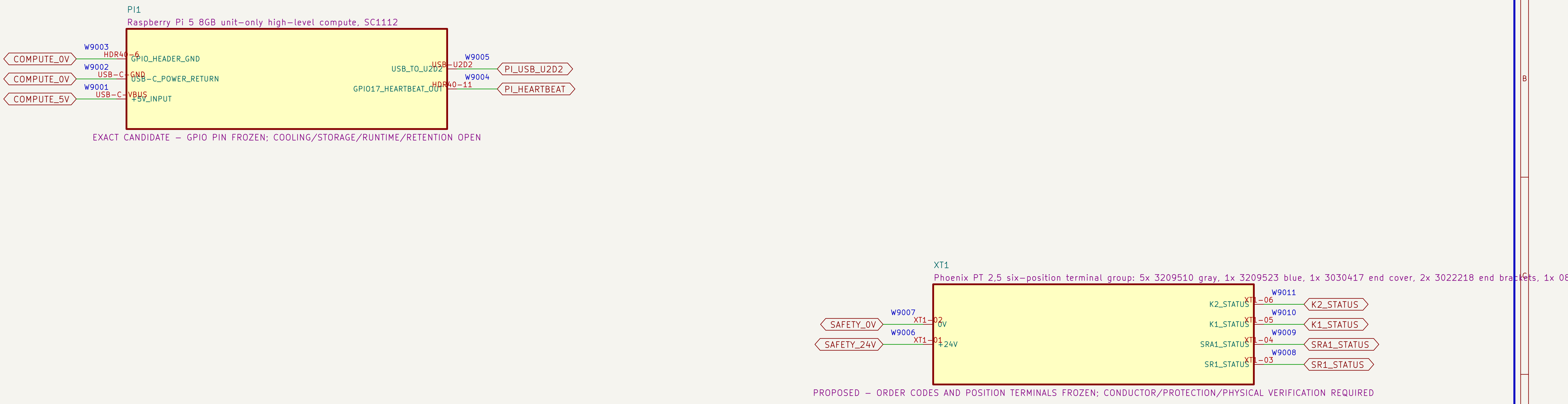
CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /08 Calculated dual–channel 24 V watchdog feedback/
File: 08_watchdog_feedback_interface.kicad_sch

Title: PB HR–V0 V3–P1.14 – 08

Size: A3 | Date: 2026–08–08 | **Rev: V3–P1.14**
KiCad E.D.A. 10.0.5 | Id: 9/13

09 Compute and control terminals

High−level compute and diagnostic wiring have no authority to bypass or restore the safety chain.



NOTE 1: No installed debug connector exists; watchdog programming uses only TP15/TP16/TP2
under a future controlled unpowered−fixture procedure.

NOTE 2: Debug connection, halt, flash and disconnect must not enable outputs, back−power the
board or bypass a protective function.

NOTE 3: Heartbeat cable, GPIO runtime, timing, terminal family, markers, conductor range,
torque and enclosure layout remain selection required.

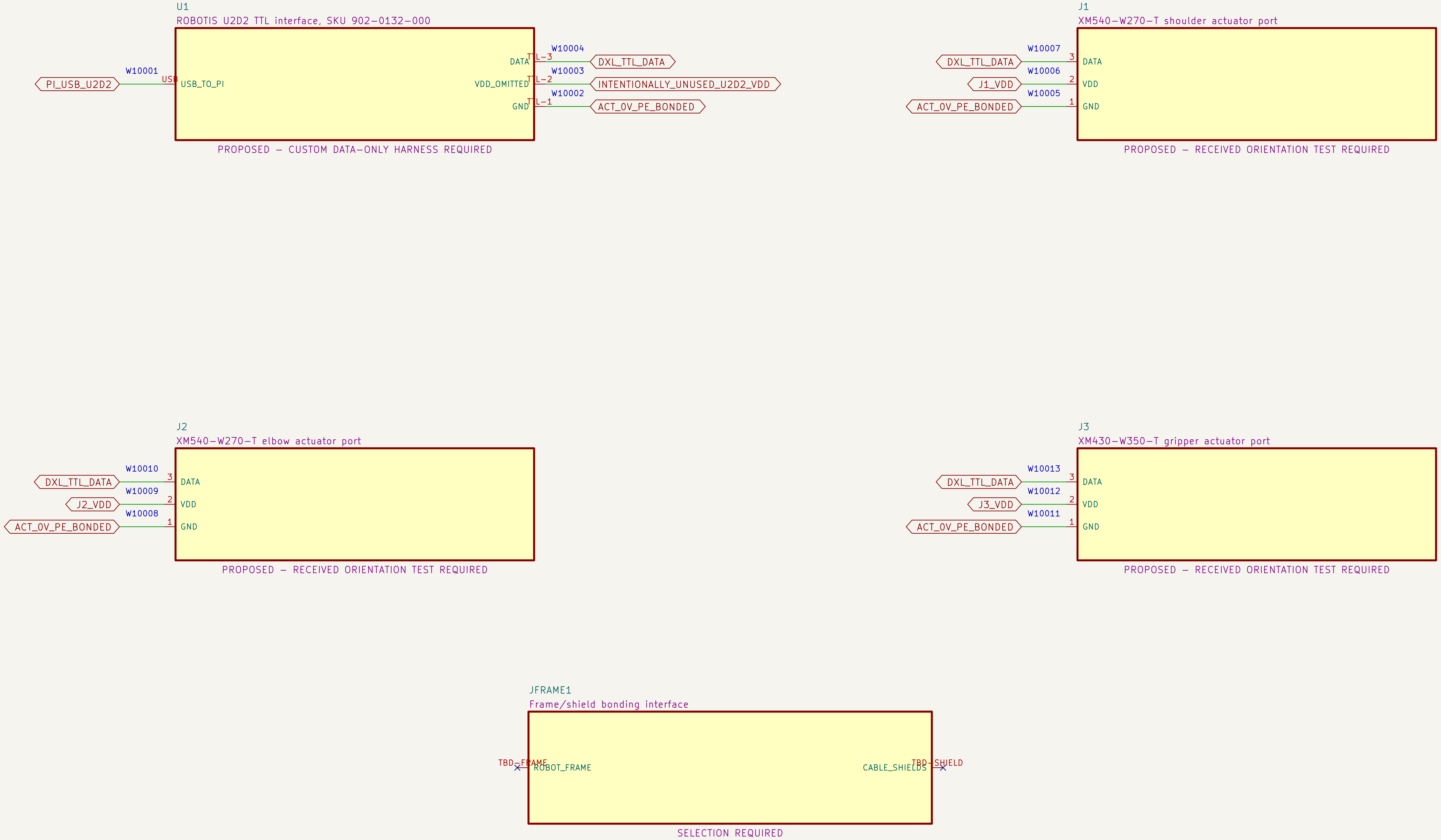
CONNECTED DESIGN CANDIDATE — QUALIFIED REVIEW REQUIRED
PRELIMINARY — NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /09 Compute and control terminals/
File: 09_compute_and_control_terminals.kicad_sch

Title: PB HR−V0 V3−P1.14 — 09

Size: A3	Date: 2026−08−08	**Rev: V3−P1.14**
KiCad E.D.A. 10.0.5		Id: 10/13

10 U2D2, actuator ports and bonding boundary

The U2D2 cable carries DATA and GND only; protected VDD is injected at each actuator.



NOTE 1: The V0 bus is TTL; HR-30 RS-485 remains a separate architecture.

NOTE 2: Do not bond frame or shields to 0V/PE until EMC and parallel-path review accepts the exact implementation.

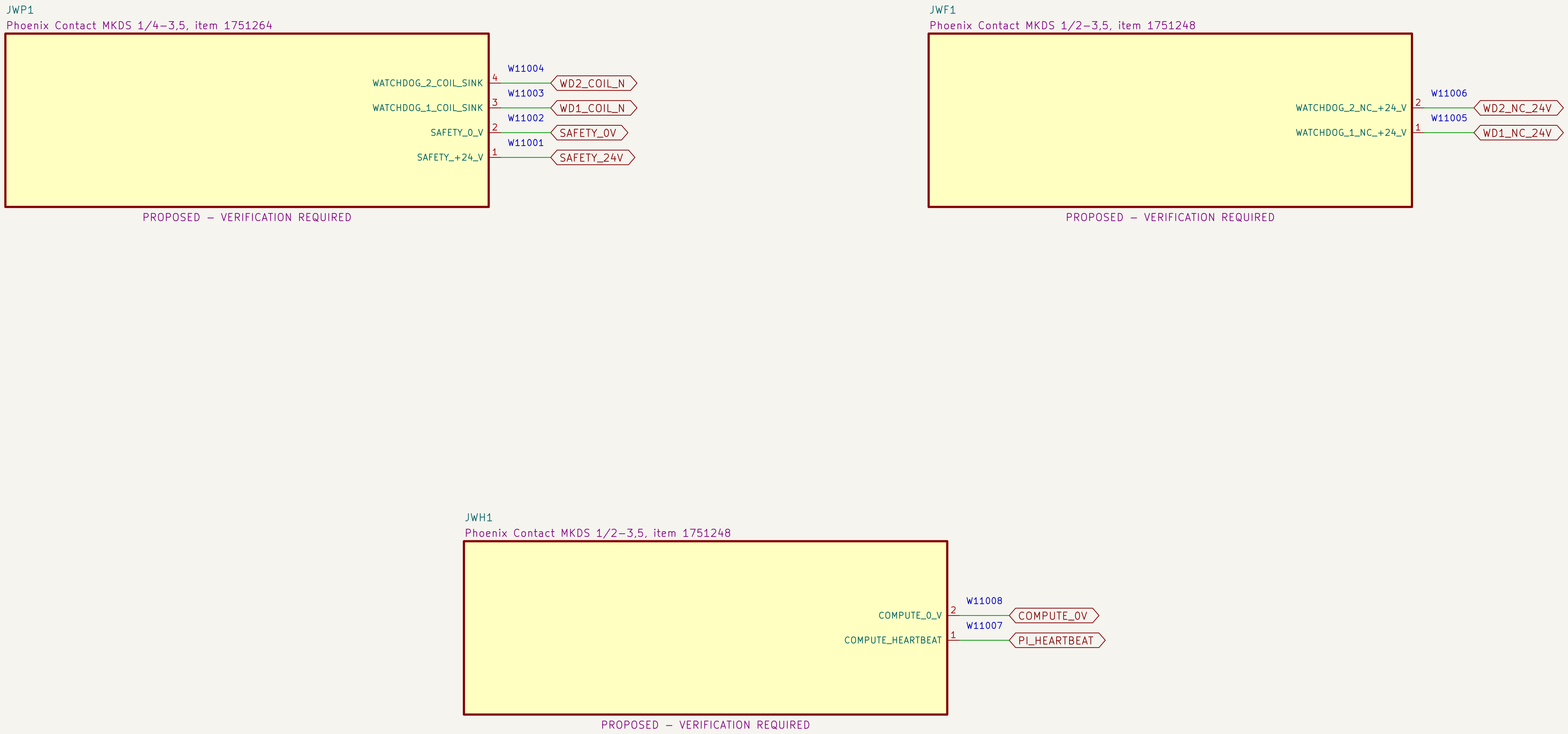
CONNECTED DESIGN CANDIDATE — QUALIFIED REVIEW REQUIRED
PRELIMINARY — NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /10 U2D2, actuator ports and bonding boundary/
File: 10_actuator_interfaces.kicad_sch

Title: PB HR-V0 V3-P1.14 — 10

Size: A3	Date: 2026-08-08	**Rev: V3-P1.14**
KiCad E.D.A. 10.0.5		Id: 11/13

11 Watchdog PCB external connectors

Exact PCB terminal-block candidates and project pin allocations define the board boundary; harness release remains open.

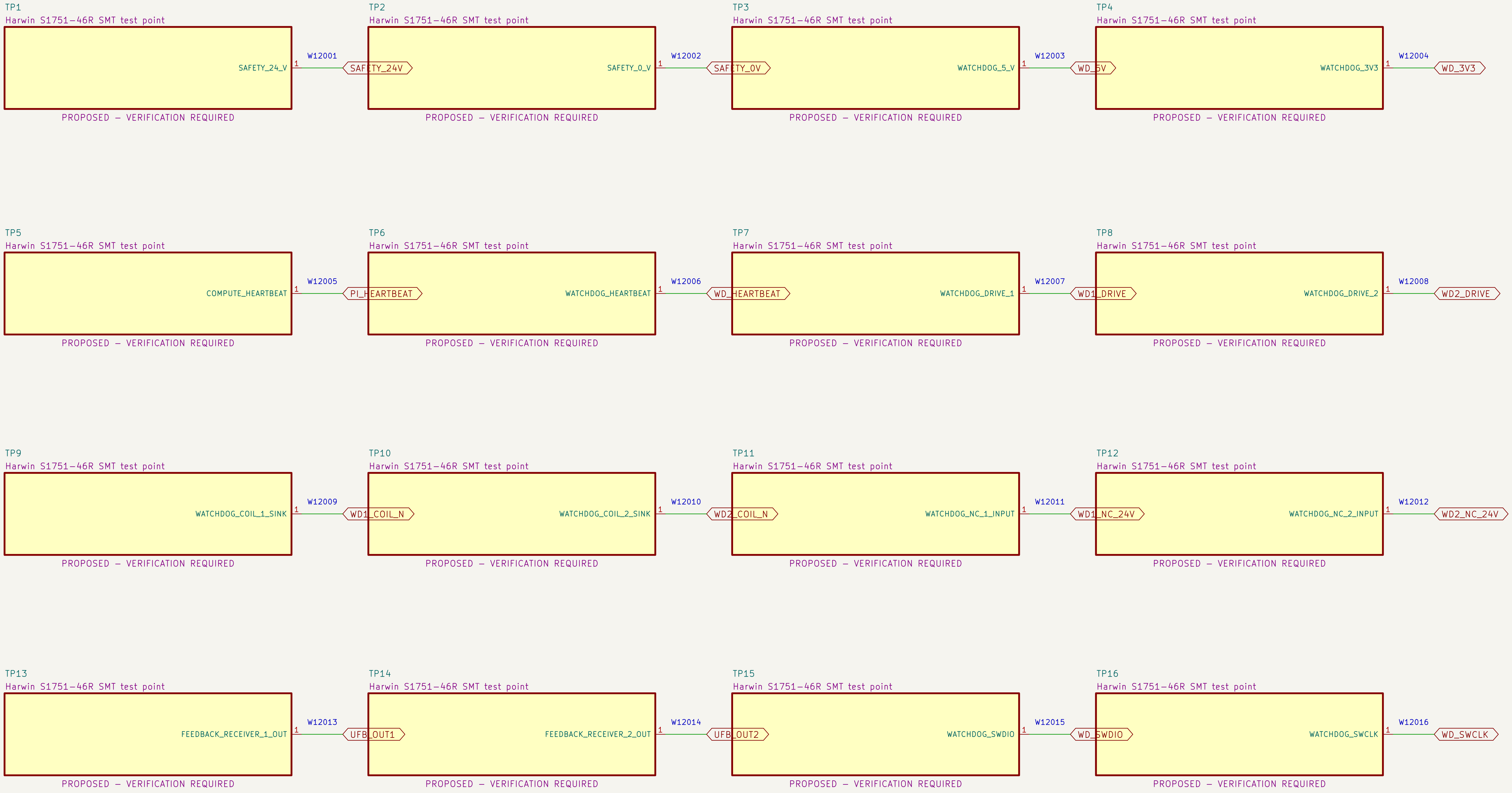


NOTE 1: Terminal numbering follows the PCB footprint and must be confirmed against received parts before wiring.

NOTE 2: Nominal terminal ratings do not release branch protection, conductor, ferrule, harness, enclosure or thermal application.

12 Watchdog PCB test access

Exact clip–compatible test terminals expose controlled nodes without granting safety or operating authority.



NOTE 1: Test terminals are diagnostic features only; they provide no safety integrity or bypass authority.

NOTE 2: Verify clip clearance with guards, harnesses and power removed before any live measurement procedure is released.

CONNECTED DESIGN CANDIDATE – QUALIFIED REVIEW REQUIRED
PRELIMINARY – NOT APPROVED FOR FABRICATION OR ENERGIZATION
Project Button
Sheet: /12 Watchdog PCB test access/
File: 12_watchdog_pcb_test_access.kicad_sch

Title: PB HR-V0 V3–P1.14 – 12

Size: A3 | Date: 2026-08-08 | **Rev: V3–P1.14**
KiCad E.D.A. 10.0.5 | Id: 13/13